

# A Packing of 70 Lines in $\mathbb{R}^{13}$

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## Abstract

We give 70 1-dimensional subspaces of  $\mathbb{R}^{13}$  whose smallest squared chordal distance is at least 0.902154. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.901374, obtained by deleting 1 of the 71 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 70 subspaces of dimension 1 in  $\mathbb{R}^{13}$  whose pairwise squared chordal distances  $D(a, b) = 1 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 0.902154\dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 13$ ,  $n = 1$ . The cell  $N = 70$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.901374, obtained by deleting 1 of the 71 lines of the packing found by this search. The Rankin simplex bound for this cell is  $1(13 - 1)/13 \cdot 70/69 = 0.936455$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N70.txt`

in <https://github.com/Ge-limin/math719>.

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\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 72 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:  
451077103136772209483956920438055723905548812449  
denominator:  
499999999999777405224916493699192209753817561411

## B Coordinates

70 blocks of 13 rows with 1 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m13n1N70.txt`.

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